MATH-361 Probability and Statistics

Normal Distribution

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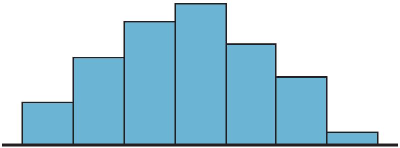
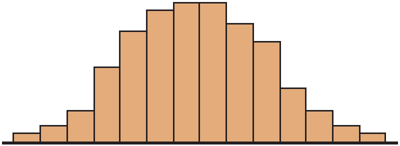
Normal Distribution

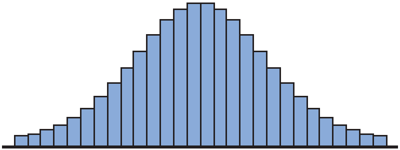
If a researcher selects a random sample of 100 adult women, measures their heights, and constructs a histogram. If the researcher increases the sample size and decreases the width of the classes, the histograms will look like the ones shown in Figures



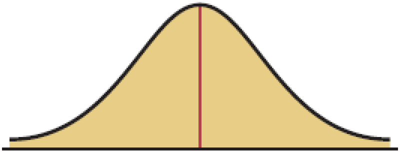
(a) Random sample of 100 women



(b) Sample size increased and class width decreased

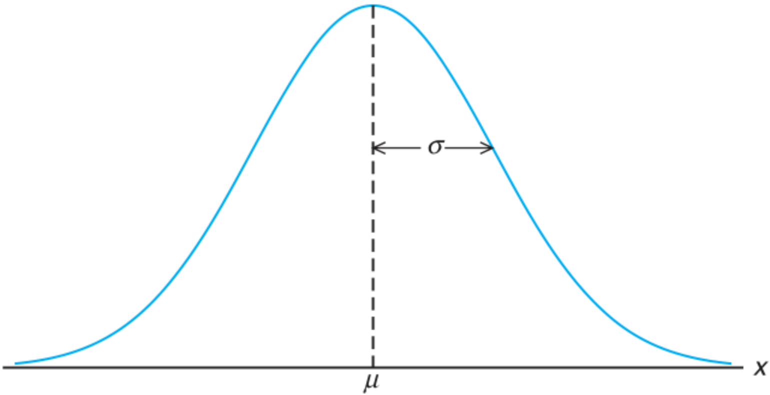


(c) Sample size increased and class width decreased further



(d) Normal distribution for the population

The histogram would approach what is called a `normal distribution`.



This distribution is also known as a **bell curve** or a **Gaussian distribution**, named for the German mathematician Carl Friedrich Gauss (1777-1855), who derived its equation.

In mathematics, curves can be represented by equations. For example, the equation of the circle is $x^2 + y^2 = r^2$, where r is the radius. A circle can be used to represent many physical objects, such as a wheel. Even though it is not possible to manufacture a wheel that is perfectly round, the equation and the properties of a circle can be used to study many aspects of the wheel, such as area, velocity, and acceleration. In a similar manner, the theoretical curve, called a normal distribution curve, can be used to study many variables that are not perfectly normally distributed but are nevertheless approximately normal.

We say that X is a normal random variable, or simply that X is normally distributed, with parameters μ and σ^2 if the density of X is given by

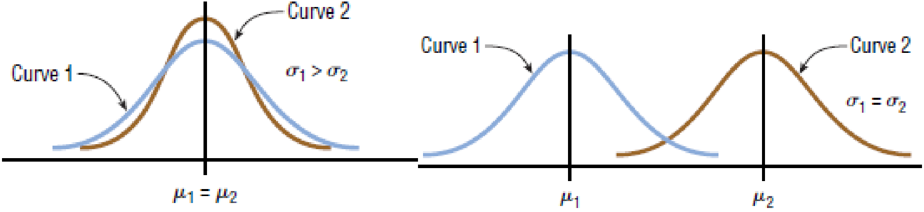
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad -\infty < x < \infty$$

where

$e \approx 2.718 \quad \pi \approx 3.14$

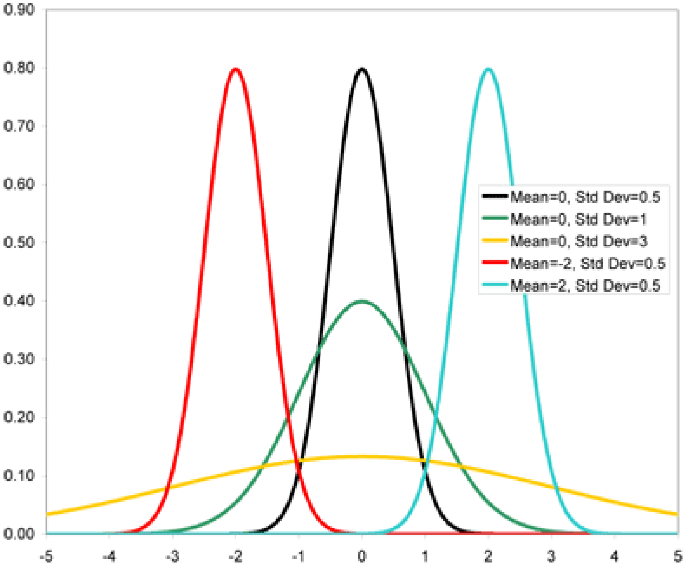
$\mu \quad = \text{Population mean}$

$\sigma \quad = \text{Population standard deviation.}$

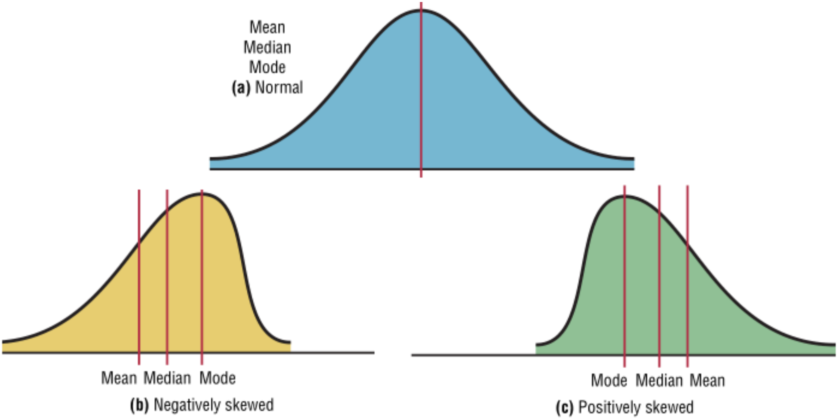


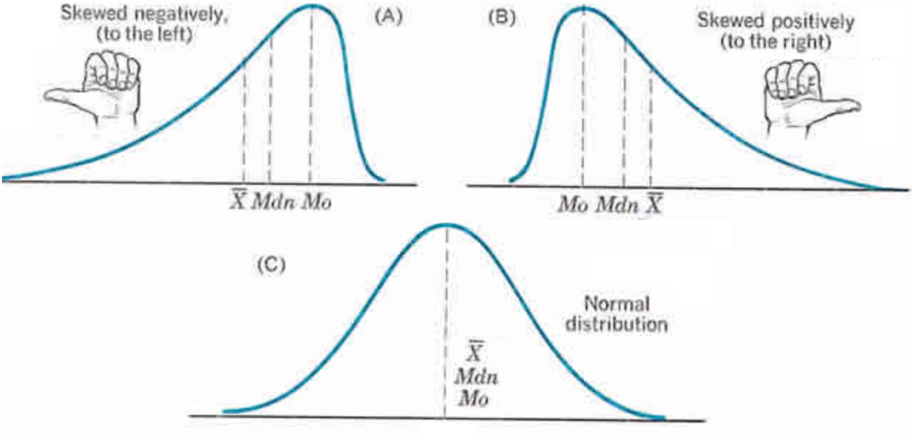
(a) Same means but different standard deviations

(b) Different means but same standard deviations


0.90
0.80
0.70
0.60
0.50
0.40
0.30
0.20
0.10
0.00
-5
-4
-3
-2
-1
0
1
2
3
4
5
Mean=0, Std Dev=0.5
Mean=0, Std Dev=1
Mean=0, Std Dev=3
Mean=-2, Std Dev=0.5
Mean=2, Std Dev=0.5

No variable fits a normal distribution perfectly, since a normal distribution is a theoretical distribution. However, a normal distribution can be used to describe many variables, because the deviations from a normal distribution are very small.




Skewed negatively,
(to the left)
(A)
$\bar{X}$ Mdn Mo
(B)
Skewed positively
(to the right)
Mo Mdn $\bar{X}$
(C)
Normal
distribution
$\bar{X}$
Mdn
Mo

1. A normal distribution curve is **bell-shaped**.
2. The **mean**, **median**, and **mode** are **equal** and are located at the center of the distribution.
3. A normal distribution curve is **unimodal** (it has only one mode).
4. The curve is **symmetric** about the mean, which is equivalent to saying that its shape is the same on both sides of a vertical line passing through the center.
5. The curve is **continuous**; that is, there are no gaps or holes in the curve.
6. The curve **never touches the x-axis**. Theoretically, no matter how far in either direction the curve extends, it never meets the x axis but it gets increasingly closer.

7. **The total area under a normal distribution curve is equal to 1.00, or 100%.**

To prove that $f(x)$ is a probability density function, showing that

$$\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\,dx = 1$$

$$\frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \overbrace{=}^{\boxed{y=\frac{x-\mu}{\sigma}}} \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy = \frac{1}{\sqrt{2\pi}} I$$

Now $I^2 = \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy . \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} dxdy$

$$x = r\cos\theta, \quad y = r\sin\theta \quad \Longrightarrow \quad dxdy = rd\theta dr$$

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} dxdy = \int_{0}^{\infty}\int_{0}^{2\pi} e^{-\frac{r^2}{2}} rd\theta dr = 2\pi$$

8. The **approximate area** under the part of a normal curve that lies within

1 SD	$\mu \pm \sigma$	68%
2 SD	$\mu \pm 2\sigma$	95%
3 SD	$\mu \pm 3\sigma$	99.7%

"SD" stands for

Standard Deviation

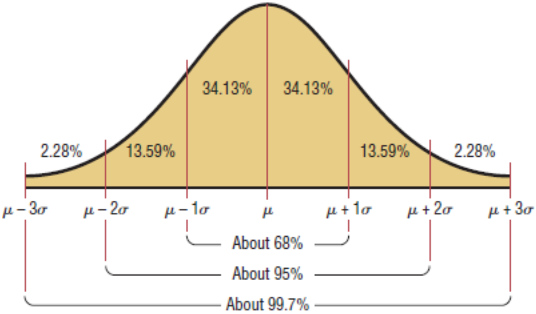


Figure: Area under normal distribution curve.

Areas under the Normal Curve

The area under the **normal curve** bounded by the two ordinates $x = x_1$ and $x = x_2$ equals the probability that the random variable X assumes a value between $x = x_1$ and $x = x_2$. Thus, for the normal curve in Figure

$$\text{Areas under the Normal Curve} = P(x_1 < X < x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

is represented by the area of the shaded region

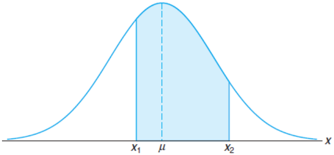


Figure: $P(x_1 < X < x_2)$ = area of the shaded region

Standard Normal Distribution

The distribution of a normal random variable with **mean 0** and **variance 1** is called a **standard normal distribution**.

If we let Z be the random variable corresponding to the following

$$Z = \frac{X-\mu}{\sigma}$$ Z is called the **standard variable** corresponding to X

The mean or expected value of Z is 0 and the standard deviation is 1. In such cases the density function for Z can be obtained from the definition of a normal distribution by allowing $\mu = 0$ and $\sigma = 1$, yielding

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \boxed{Z = \frac{X-\mu}{\sigma}} \implies f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

A Professor is marking a test. Here are the student's results (out of 60 points):

$$20,\ 15,\ 26,\ 32,\ 18,\ 28,\ 35,\ 14,\ 26,\ 22,\ 17$$

Most students didn't even get 30 out of 60, and **most will fail**.
The test must have been really hard, so the Prof decides to Standardize all the scores and only fail people 1 standard deviation below the mean.
The Mean is 23, and the Standard Deviation is 6.6, and these are the Standard Scores:

$$-0.45,\ -1.21,\ 0.45,\ 1.36,\ -0.76,\ 0.76,\ 1.82,\ -1.36,\ 0.45,\ -0.15,\ -0.91$$

Now only 2 students will fail (the ones lower than -1 standard deviation)

A survey of daily travel time had these results (in minutes):

$$26, 33, 65, 28, 34, 55, 25, 44, 50, 36, 26, 37, 43, 62, 35, 38, 45, 32, 28, 34$$

The **Mean is 38.8** minutes, and the **Standard Deviation is 11.4** minutes. **Convert the values to z-scores ("standard scores").**

Original Value	$z = \frac{x-u}{\sigma}$	Standard Score
26	$\boxed{\frac{26-38.8}{11.4}}$	-1.12
33	$\boxed{\frac{33-38.8}{11.4}}$	-0.51
65	$\boxed{\frac{65-38.8}{11.4}}$	2.30
...	...	...

Normal Distribution Table

Normal Distribution Table gives the area under the normal distribution curve to the **left of any z value given** in two decimal places.

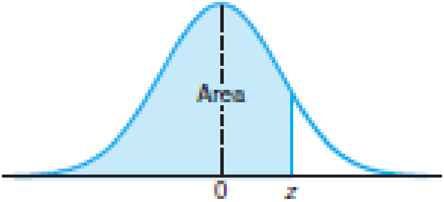


Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Table A.3 (continued) Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

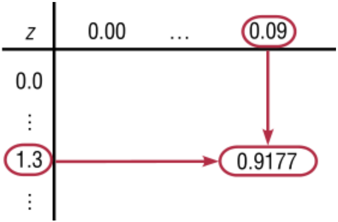
Normal Distribution Table

Table gives the area under the normal distribution curve to the left of any z value given in two decimal places.

For example, the area to the left of a z value of 1.39 is found by looking up

1. Right of 1.3
2. Below 0.09

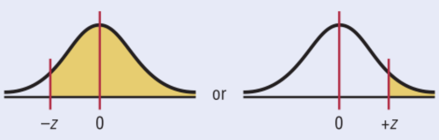
Where the two lines meet gives an area of 0.9177.

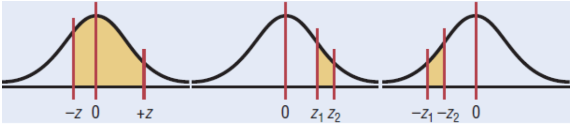


Finding Area in Standard Normal Distribution Curve

There might be 3 different types of values

1. To the left of z.
2. To the right of z.
3. Between any Two value of z





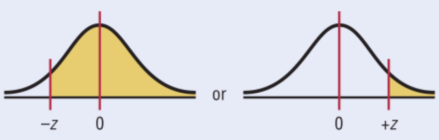
Finding Area in Standard Normal Distribution Curve

To the left of any z value:

Look up the z value in the table and use the area given.

To the right of any z value:

Look up the z value in the table and subtract the value of z from 1.



Finding Area in Standard Normal Distribution Curve

To the left of any z value:

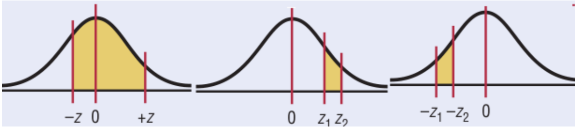
Look up the z value in the table and use the area given.

To the right of any z value:

Look up the z value in the table and subtract the value of z from 1.

Between any two z values:

Look up both z values and subtract the corresponding areas.

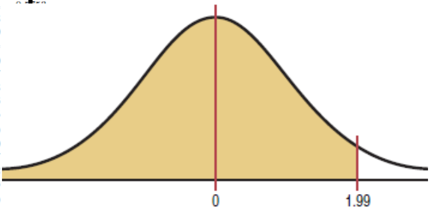


Find the area to the left of z = 1.99.

The area to the left of a z value of 1.99 is found by looking up 1.9 in the left column and 0.09 in the top row. It is 0.9767. Hence 97.67% of the area is less than z =1.99.

Table A.3 (continued) Areas under the Normal Curve

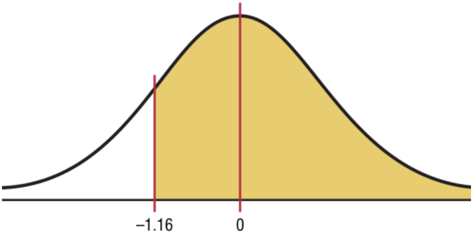
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	[illegible]
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0.9767



Find the area to the right of z =-1.16.

Table A.3 Areas under the Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	[illegible]	0.0244
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	[illegible]	0.0307
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	[illegible]	0.0384
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	[illegible]	0.0475
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	[illegible]	0.0582
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	[illegible]	0.0708
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	[illegible]	0.0853
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	[illegible]	0.1020
−1.1	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	0.1230	[illegible]
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	[illegible]	0.1423



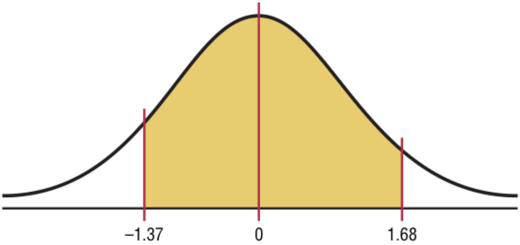
1. Look up the area for z =-1.16. It is 0.1230.
2. Subtract it from 1. $1.000 - 0.1230 = 0.8770$.

Hence 87.70% of the area under the standard normal distribution curve is to the left of z = -1.16.

Find the area between z=+1.68 and z=-1.37.

Since the area desired is between two given z values, look up the areas corresponding to the two z values and subtract the smaller area from the larger area.

(Do not subtract the z values.)

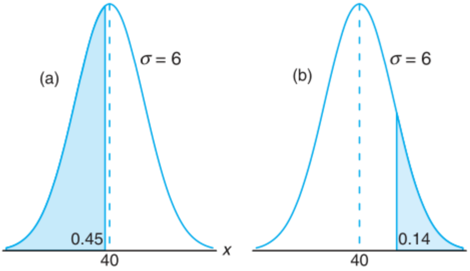


1. The area for $z = +1.68$ is 0.9535, and the area for $z = -1.37$ is 0.0853.
2. The area between the two z values is

$$0.9535 - 0.0853 = 0.8682 \text{ or } 86.82\%$$

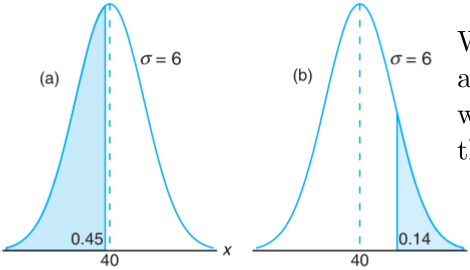
Given a normal distribution with $\mu = 40$ and $\sigma = 6$, find the value of x that has

1. 45% of the area to the left and
2. 14% of the area to the right.



Given a normal distribution with $\mu = 40$ and $\sigma = 6$, find the value of x that has

1. 45% of the area to the left and
2. 14% of the area to the right.



We require a z value that leaves an area of 0.45 to the left. From Table we find $P(Z < -0.13) = 0.45$, so the desired z value is -0.13. Hence,

$$x = (6)(-0.13) + 40 = 39.22.$$

Given a normal distribution with $\mu = 40$ and $\sigma = 6$, find the value of x that has

1. 45% of the area to the left and
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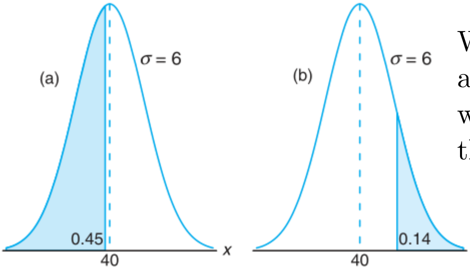

We require a z value that leaves an area of 0.45 to the left. From Table we find $P(Z < -0.13) = 0.45$, so the desired z value is -0.13. Hence,

$$x = (6)(-0.13) + 40 = 39.22.$$

This time we require a z value that leaves 0.14 of the area to the right and hence an area of 0.86 to the left. Again, from Table, we find $P(Z < 1.08) = 0.86$, so the desired z value is 1.08 and $x = (6)(1.08) + 40 = 46.48$.

A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find the probability that a given battery will last less than 2.3 years.

A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find the probability that a given battery will last less than 2.3 years.

$$\mu = 3, \quad \sigma = 5, \quad z = \frac{x-\mu}{\sigma} = -1.4, P(z < -1.4) + ?$$

$$Z = \frac{2.3-3}{0.5} = -1.4; P(X < 2.3) = P(Z < -1.4) = 0.0808.$$

A certain type of storage battery lasts, on average, 3.0 years with a standard deviation of 0.5 year. Assuming that battery life is normally distributed, find the probability that a given battery will last less than 2.3 years.

$$\mu = 3, \quad \sigma = 5, \quad z = \frac{x - \mu}{\sigma} = -1.4, P(z < -1.4) + ?$$

An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

The distribution of light bulb life is illustrated in Figure. The z values corresponding to $x_1 = 778$ and $x_2 = 834$ are

$$z_1 = -0.55, \quad z_2 = 0.85, \quad P(-0.55 \leq x \leq 0.85)$$